Homework 7

Week 7

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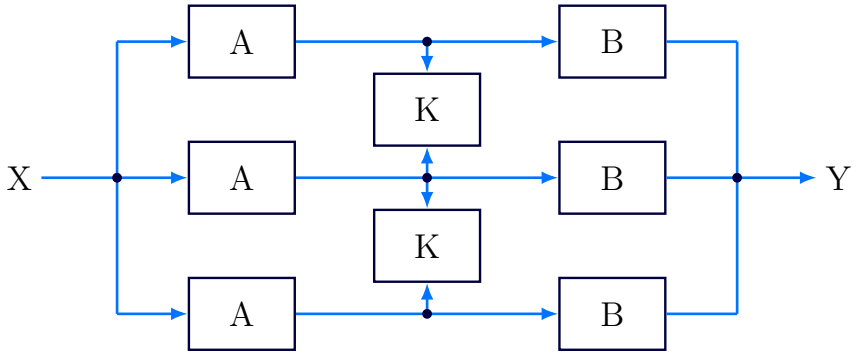
Problem 1. A system which is operated from X to Y via the redundant non serial-parallel path as shown in the figure below. Let each module (A, B, and K) has their reliabilities (R_a, R_b, and R_k) respectively and ignore the reliabilities of the connections.

Find the system reliability.



Solution

Let m_1, m_2, m_3 be the midpoints of the three $A \to B$ paths (top to bottom). K_1 bridges m_1–m_2 and K_2 bridges m_2–m_3, so the network is not series-parallel. Use the decomposition (conditional probability) method on the K modules: X reaches Y iff some working A_i and some working B_j have m_i, m_j joined by working K's. Define

$$P = R_a R_b, \quad Q_2 = \left[1-(1-R_a)^2\right]\left[1-(1-R_b)^2\right], \quad Q_3 = \left[1-(1-R_a)^3\right]\left[1-(1-R_b)^3\right].$$

- K_1, K_2 **both good** (R_k^2): m_1, m_2, m_3 merge into one node, giving 3 parallel A's in series with 3 parallel B's: $R = Q_3$.
- **Exactly one** K **good** ($2R_k(1-R_k)$, symmetric): two paths merge into a "2 A ‖ – 2 B ‖" block, in parallel with the remaining single A–B path: $R = 1-(1-Q_2)(1-P)$.
- **Both** K **failed** ($(1-R_k)^2$): three independent A–B paths in parallel: $R = 1-(1-P)^3$.

By the total probability theorem,

$$\boxed{R_{sys} = R_k^2\, Q_3 + 2R_k(1-R_k)\left[1-(1-Q_2)(1-R_aR_b)\right] + (1-R_k)^2\left[1-(1-R_aR_b)^3\right]}$$

Check: $R_k = 1$ gives Q_3 (fully cross-connected) and $R_k = 0$ gives three plain parallel paths, as expected.